Adverse Childhood Experiences and Chronic Pain in U.S. Children

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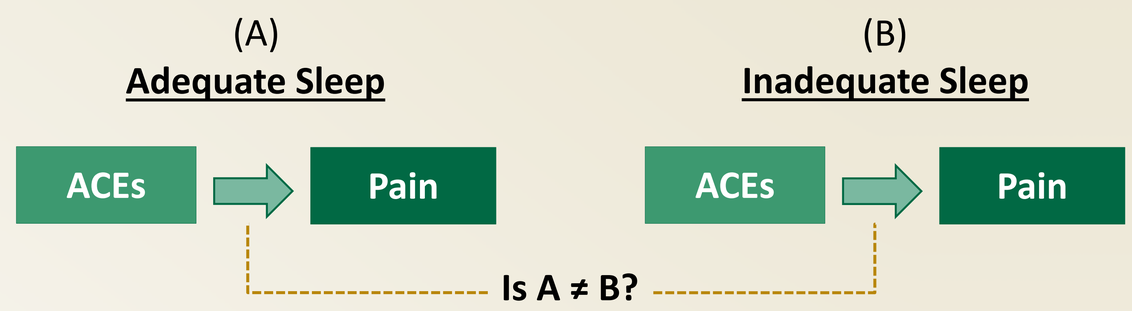
Background and Objectives

- Adverse childhood experiences (ACEs) affect more than 1 in 3 U.S. children.
- ACEs are strongly associated with chronic pain across the lifespan.
- Sleep is associated with stress physiology and pain outcomes, but it is unknown whether sleep quality modifies ACE-pain associations.
- Identifying modifiable protective factors is essential for designing effective pediatric health interventions.
- This study examines whether adequate sleep modifies the association between ACE exposure and chronic or recurrent pain in a nationally representative sample of children.



Methods

Data Source and Sample
- National Survey of Children's Health, 2022-2023
- Children aged 6-17 years
- Analytic sample: n = 80,874

Measures
Outcome: Chronic pain
Exposure: 10 ACE types; cumulative ACE counts (0, 1, 2, 3+)
Modifier: Sleep adequacy (age-specific)

Covariates
1. Age
2. Sex
3. Race/ethnicity
4. Poverty Level
5. Parental mental health
6. Special healthcare needs status

Analysis
- Survey-weighted logistic regression
- Effect modification evaluated via interaction terms and Wald F-test
- Stratified models by sleep status

Results

Dose-response association with cumulative ACEs

- Cumulative ACE exposure showed a dose-response association with chronic or recurrent pain.
- After adjustment for sociodemographic and health-related covariates, children with three or more ACEs had more than twice the odds of chronic pain compared with children with no ACE exposure.

Table 1: Adjusted odds of chronic pain by cumulative ACE exposure

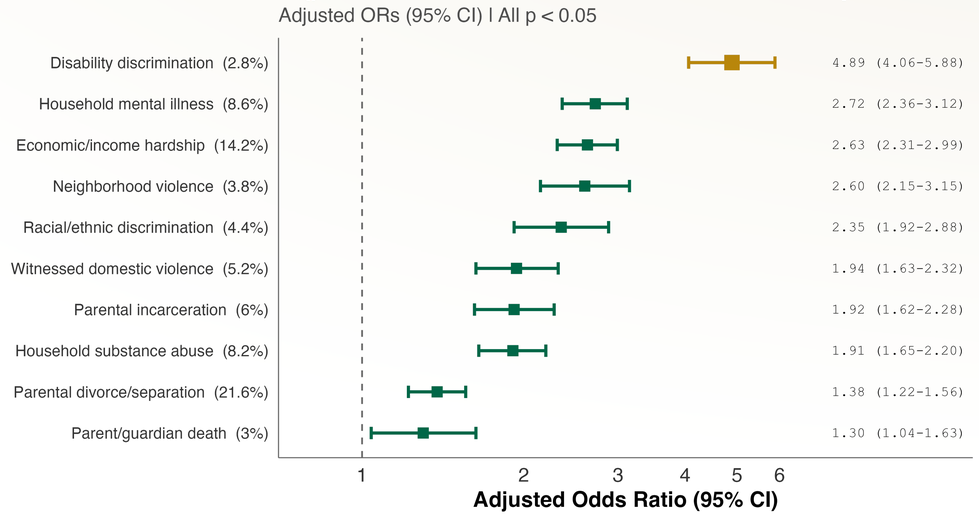
Cumulative ACEs	Adjusted OR	95% CI
0 (Ref)	**1.00**	–
1 ACE	**1.54**	(1.30–1.84)
2 ACEs	**1.91**	(1.54–2.38)
3+ ACEs	**2.15**	(1.75–2.64)

Adjusted for age, sex, race/ethnicity, poverty level, parental mental health, and special healthcare needs status.

Heterogeneity in ACE-pain associations

- All 10 ACE types were significantly associated with pain (all $p < 0.05$).
- Disability discrimination showed the strongest association (OR = 4.89), followed by parental mental illness (OR = 2.72) and economic hardship (OR = 2.63).
- Effect sizes ranged from OR = 1.30 (parental/guardian death) to OR = 4.89 (disability discrimination).

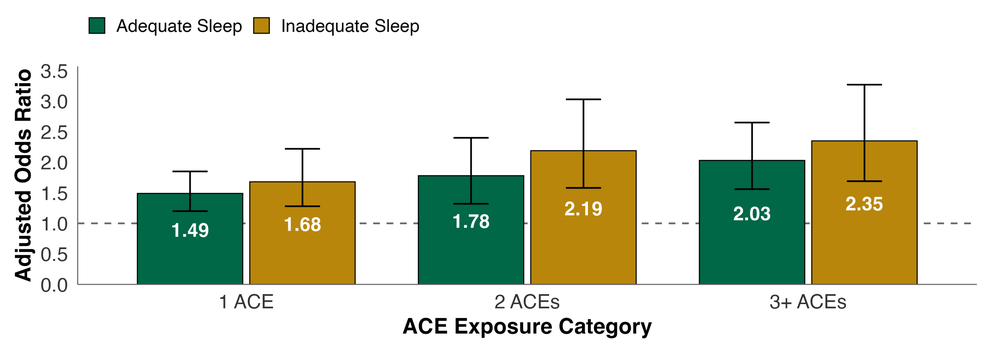
Figure 1: Individual ACE associations with pain
Adjusted ORs (95% CI) | All $p < 0.05$



No evidence of effect modification by sleep quality

- Sleep did not modify the ACE association with pain (interaction test $p = 0.605$).
- ACE-pain associations were similar across adequate and inadequate sleepers.
- Stratified analysis confirmed similar ACE-pain associations across adequate and inadequate sleepers at all ACE exposure levels.

Figure 2: ACE-pain associations by sleep status
Interaction test: $p = 0.605$



Conclusion

Adverse childhood experiences (ACEs) demonstrate a dose-response relationship with childhood chronic or recurrent pain. Disability discrimination shows the strongest pain association of all ACEs. While sleep is a protective health factor, sleep adequacy did not modify the ACE-pain association in children between the ages of 6 to 17. These findings suggest sleep promotion alone may not be an appropriate lifestyle intervention to reduce ACE-associated pain burdens in children. Future research should investigate other modifiable protective health factors for their potential to buffer ACE-pain associations.

Acknowledgements
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